FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

HIV/AIDS in Pregnancy

Lecture | Week 6 · Obstruction, HIV, diabetes & the written test | Obstetrics | Dr Nsingo

Every pregnant woman in Uganda is offered HIV, syphilis and hepatitis B testing at her first ANC visit; every woman with HIV starts lifelong TLD the same day, has her viral load checked every 3 months until breastfeeding ends, and her baby gets risk-based prophylaxis (NVP alone, or AZT + NVP if high-risk) plus DNA PCR testing at 6 weeks, 9 months and 6 weeks after weaning.

Must know before the ward round

1. Without any intervention MTCT is 15–45%; with a suppressed maternal viral load throughout pregnancy and breastfeeding it is below 1%. Viral load is the single biggest determinant of transmission.
2. Uganda uses test-and-treat (Option B+, since 2012) for all pregnant and breastfeeding women; start TLD (TDF 300 mg / 3TC 300 mg / DTG 50 mg) once daily, ideally the same day as diagnosis.
3. Viral load at the first ANC visit (if already on ART) or 3 months after starting ART, then every 3 months until breastfeeding stops. VL of 1,000 copies/mL or more = non-suppressed = high-risk infant.
4. Four facts to present for every HIV-positive mother are regimen, time on ART, last VL (result AND date) and adherence. An unknown or old VL near term means you treat the baby as high-risk.
5. Low-risk infant = NVP once daily for 6 weeks. High-risk infant = AZT + NVP for 6 weeks, then NVP alone for another 6 weeks (12 weeks total), extended if the mother is still not suppressed.
6. Start infant prophylaxis within 6 hours of birth; NVP and AZT are both 15 mg (1.5 mL of 10 mg/mL syrup) for a baby of 2.5 kg or more (10 mg if 2.0–2.49 kg).
7. EID = DNA PCR (DBS or point-of-care) at 4–6 weeks, at 9 months and 6 weeks after complete cessation of breastfeeding, then a rapid antibody test at 18 months.
8. Exclusive breastfeeding for 6 months, then complementary foods with continued breastfeeding to at least 12 months, as long as the mother is on ART. Never mix-feed in the first 6 months.
9. Co-trimoxazole 960 mg once daily for HIV-positive pregnant and breastfeeding women in Uganda, so NO IPTp-SP (both are sulfa drugs). Separate DTG from iron/calcium/antacids (2 h before or 6 h after), and give DTG 50 mg twice daily with rifampicin.
10. Advanced HIV disease (CD4 below 200 or WHO stage 3–4) triggers serum CrAg and urine TB LAM; if cryptococcal meningitis, delay ART 4–6 weeks after starting antifungals.
11. Triple elimination at booking means syphilis-positive women get benzathine penicillin 2.4 MIU IM (weekly × 3 if duration unknown) and HBsAg-positive women get TDF from about 28 weeks plus a birth-dose HBV vaccine for the baby within 24 hours.
12. Caesarean section is NOT routinely indicated for HIV in Uganda; do it for obstetric indications. In labour, avoid prolonged rupture of membranes, unnecessary ARM, episiotomy and scalp electrodes.

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1. Definition

HIV in pregnancy is an eMTCT problem (elimination of mother-to-child transmission). The virus can pass to the baby **in utero, during labour and delivery, and through breastfeeding**. The whole strategy boils down to one idea: **get the mother's viral load undetectable as early as possible and keep it there until breastfeeding ends**, and cover the baby with prophylaxis during the window of highest risk.

In Uganda that means:

1. **Test** every pregnant woman at the first ANC visit (HIV/syphilis dual test plus HBsAg: the **triple elimination** package) and **retest** those who are negative.
2. **Treat** every woman with HIV with lifelong **TLD** the **same day** (test-and-treat, formerly called Option B+).
3. **Monitor** viral load every 3 months through pregnancy and breastfeeding.
4. **Protect the baby**: risk-based prophylaxis, early infant diagnosis (EID), co-trimoxazole and safe infant feeding.

Term	Meaning
MTCT / vertical transmission	Transmission of HIV from mother to child during pregnancy, labour/delivery or breastfeeding
eMTCT	Uganda's programme to eliminate MTCT; the target is fewer than 50 new paediatric infections per 100,000 live births and a final transmission rate below 5% in breastfeeding populations
Triple elimination	Integrated elimination of vertical transmission of **HIV, syphilis and hepatitis B**
Option B+ / test-and-treat	Lifelong ART for every pregnant or breastfeeding woman with HIV, regardless of CD4 or clinical stage
HIV-exposed infant (HEI)	Any infant born to a mother living with HIV, until HIV infection is excluded after breastfeeding ends
Viral suppression	VL below 1,000 copies/mL (the programme threshold). **Undetectable** = below the assay's detection limit (usually below 50 copies/mL). **Low-level viraemia** = detectable but below 1,000 copies/mL
Non-suppressed / unsuppressed	VL of 1,000 copies/mL or more; triggers intensive adherence counselling (IAC) and a repeat VL
High-risk HEI	Infant at raised risk of infection (section 5); gets enhanced prophylaxis (AZT + NVP)
Advanced HIV disease (AHD)	CD4 below 200 cells/mm³ or WHO clinical stage 3 or 4
EID	Early infant diagnosis by DNA PCR, because maternal antibodies make antibody tests unreliable until about 18 months

CLINICAL PEARL

The one thing never to forget: **the infant's risk is set by the mother's most recent viral load.** Before every delivery of an HIV-positive woman, ask "What was her last viral load, and when?" The answer decides whether the baby gets NVP alone or AZT + NVP.

2. Epidemiology

- **Transmission without and with intervention**: without any intervention MTCT is **15–45%**; with a suppressed maternal viral load throughout pregnancy and breastfeeding it is **below 1%**.

- Uganda has one of the largest HIV epidemics in East Africa, with adult prevalence of roughly **5–6%**, and it is **higher in women** than in men Verify *figure against the latest UPHIA/UNAIDS estimate*. Young women aged 15–24 carry a disproportionate share of new infections, which is exactly the age group filling our antenatal clinics.
- Uganda was one of the first countries to adopt **Option B+ in 2012**. Coverage of ART among HIV-positive pregnant women is now very high, and Uganda has recorded a large reduction in new infections among newborns.
- The **remaining infections** come mainly from: women who **seroconvert during pregnancy or breastfeeding** (they tested negative at booking); women who **disengage from care** (especially postpartum); **non-suppressed** women with poor adherence; and **missed EID** so infected infants are found late.
- HIV services are integrated in ANC at HC III and above. **Point-of-care (POC) viral load and POC EID** machines are available at many HC IVs and hospitals; elsewhere samples go by the national **hub transport system** to CPHL, and results can take weeks. That delay is why you must plan ahead: order the VL early.
- **Dual HIV/syphilis rapid tests** have dramatically increased syphilis screening in ANC; **HBsAg testing** is being scaled up but is less consistently available.

> **UGANDA CONTEXT**
> In Uganda the HIV-positive mother is usually managed in the **eMTCT/ART clinic** within the ANC, with a **"mother-baby pair" follow-up** until the baby's final HIV status is known. As a clerk, check her **ART card / HMIS register**: regimen, start date, last VL and date, adherence and TB screening are recorded there.

3. Aetiology and Risk Factors

Cause: infection with the human immunodeficiency virus (in Uganda almost all HIV-1), a retrovirus that infects CD4 T cells. Women acquire it mainly through heterosexual sex; the baby acquires it from the mother.

Risk factors for a woman acquiring HIV in pregnancy or breastfeeding (these are the women to offer PrEP and to retest)

- HIV-positive partner who is **not suppressed or of unknown status** (serodiscordant couple).
- Multiple partners, sex work, a **recent STI**.
- Partner at high risk (e.g. fishing communities, truck drivers).
- Young age (15–24 years).

Risk factors for transmission to the baby

Factor	Why it raises risk
High maternal viral load (not on ART, non-suppressed, **acute seroconversion**)	More virus in blood, genital secretions and breast milk; the single biggest determinant
Late ART start (less than 4 weeks before delivery)	No time to suppress
Advanced HIV disease	High VL, opportunistic infections
Placental inflammation: malaria, chorioamnionitis	Breaks the placental barrier (in utero transmission)
Prolonged rupture of membranes, prolonged labour, ARM, episiotomy, scalp electrodes, instrumental delivery	Longer and greater exposure to maternal blood and secretions
Genital infections (ulcers, syphilis, herpes, discharge)	Raise genital VL
Mixed feeding in the first 6 months	Damages the infant gut barrier
Mastitis, cracked nipples, infant oral thrush	Raise breast-milk virus and breach the infant mucosa
Disengagement from care postpartum	VL rebounds while the baby is still breastfeeding

Causes of a non-suppressed viral load (the commonest problem on the ward): poor adherence (most common), recent ART start, drug interactions (rifampicin, anticonvulsants, iron/calcium/antacids taken with DTG), vomiting or malabsorption (hyperemesis, diarrhoea), and, least often on DTG, drug resistance (section 7).

4. Pathophysiology

How, and when, the baby gets infected

Timing	Approx. share of infections without intervention	Mechanism	What reduces it
In utero	5–10%	Transplacental passage, especially with high maternal VL, placental inflammation (malaria, chorioamnionitis)	Early ART, VL suppression, malaria prevention
Intrapartum	10–20% (the largest single window)	Baby exposed to maternal blood and cervicovaginal secretions; microtransfusions during contractions; swallowing secretions	Suppressed VL, avoiding prolonged ROM, invasive procedures, episiotomy; infant prophylaxis
Breastfeeding	5–20% over 18–24 months	Cell-free and cell-associated virus in breast milk; risk higher with mastitis, cracked nipples, oral thrush in baby, **mixed feeding**	Maternal ART with suppression, exclusive breastfeeding, infant NVP

Why viral load matters so much

The amount of virus in maternal blood and genital secretions determines exposure. Transmission risk rises steeply with VL: at undetectable VL on ART from before conception, transmission is close to zero. That is why the **timing of ART start** matters: a woman who starts ART at 12 weeks has months to suppress before delivery; a woman diagnosed at 36 weeks does not, and her baby is automatically **high-risk**.

Why "mixed feeding" is dangerous

Other foods and fluids (water, formula, porridge) in the first 6 months **damage the infant's gut mucosa** and cause inflammation, allowing HIV in breast milk to cross the gut wall. Exclusive breastfeeding keeps the gut barrier intact. That is the physiological reason behind "exclusive, never mixed."

Why DTG is preferred

Dolutegravir (DTG) is an **integrase inhibitor**. It lowers VL faster than efavirenz (important in late-presenting women), has a **high genetic barrier to resistance**, and is well tolerated. An early signal of neural tube defects with DTG at conception (Botswana, Tsepamo study) fell on further follow-up to a level very close to other ART, so WHO and Uganda now recommend DTG for **all** adults, including women who are pregnant or may become pregnant. Give folic acid as for every pregnancy.

Why pregnancy is a period of danger for the mother too

- Pregnancy itself does not accelerate HIV, but HIV increases **anaemia, malaria, TB, preterm birth, low birth weight, stillbirth and puerperal sepsis**.
- **TB** is a leading cause of death in HIV-positive pregnant women, and the classic symptom of "weight loss" is masked by pregnancy weight gain.
- Postpartum is the time women most often **drop out of care**, and VL rebounds. That rebound happens while the baby is still breastfeeding.

HIV infection: CD4 T-cell destruction	ART from early pregnancy: VL undetectable	Infant NVP or AZT + NVP, exclusive breastfeeding
↓		
High viral load in blood, genital tract, breast milk	Falling CD4: TB, PCP, cryptococcus, anaemia, malaria	MTCT below 1%
↓		
In utero: transplacental, worse with malaria or chorioamnionitis	Intrapartum: blood and secretions, prolonged ROM	Breastfeeding: worse with mixed feeding, mastitis
↓		
Infant HIV infection		

Feature links

- High maternal VL → more virus crossing the placenta and in genital secretions → in utero and intrapartum infection.
- Acute seroconversion in pregnancy → very high VL before antibodies form → very high transmission risk → retest negatives.
- Placental malaria or chorioamnionitis → placental inflammation → in utero transmission → malaria prevention matters (co-trimoxazole, ITN).
- Prolonged ROM, ARM, episiotomy, scalp electrodes → more exposure of the baby's skin and mucosa to maternal blood → intrapartum transmission.
- Mixed feeding → infant gut mucosal damage → HIV in breast milk crosses the gut → breastfeeding transmission.
- Mastitis, cracked nipples, infant thrush → more virus and breached mucosa → breastfeeding transmission.
- CD4 depletion → cell-mediated immune failure → TB, PCP, cryptococcal meningitis, oral thrush, Kaposi sarcoma, herpes zoster.
- HIV, malaria and (formerly) AZT → anaemia → pallor, fatigue, poor reserve for PPH.
- Chronic infection and opportunistic disease → poor maternal nutrition and placental function → FGR, low birth weight, preterm birth, stillbirth.
- Postpartum disengagement → ART stopped → VL rebound while breastfeeding → late infant infection.
- Starting ART in AHD → immune recovery against existing infection → IRIS (why ART is delayed in cryptococcal meningitis).
- Maternal IgG crosses the placenta → antibody test positive in every HEI up to about 18 months → infant diagnosis needs DNA PCR.

5. Classification / Staging

WHO clinical staging (summary)

Stage	Examples
1	Asymptomatic; persistent generalised lymphadenopathy
2	Weight loss below 10%, recurrent URTIs, herpes zoster, oral ulcers, papular pruritic eruptions
3	Weight loss above 10%, chronic diarrhoea or unexplained fever for more than 1 month, oral candida, **pulmonary TB**, severe bacterial infections

Stage	Examples
4	HIV wasting, PCP, **extrapulmonary TB**, cryptococcal meningitis, Kaposi sarcoma, toxoplasmosis, oesophageal candida

Advanced HIV disease = CD4 below 200 cells/mm³ or WHO stage 3 or 4.

Viral load categories

VL (copies/mL)	Category	Action
Below 50	Undetectable	Continue; repeat every 3 months in pregnancy and breastfeeding
50 to 999	Low-level viraemia (suppressed by programme definition)	Adherence support
1,000 or more	**Non-suppressed**	IAC ×3, check interactions, repeat VL; baby high-risk if near delivery

Infant risk category

High-risk HEI: an infant whose mother

- was **diagnosed in late pregnancy, in labour, or during breastfeeding**; or
- had **less than 4 weeks of ART** before delivery; or
- had a **VL of 1,000 copies/mL or more** in the last 4 weeks before delivery (or **no VL result** available near delivery); or
- **seroconverted** during pregnancy or breastfeeding.

All other HEIs are **low-risk**.

6. Clinical Features

Most HIV-positive pregnant women are **well**. The clinical task is to find the opportunistic infection, the drug problem and the non-suppression, and to classify the baby's risk.

History

Each item has a reason: if the answer changes nothing, you do not need to ask it.

Question	Why it matters
When were you diagnosed with HIV? New today, earlier this pregnancy, or years ago?	New diagnosis needs counselling, disclosure support, same-day ART and baseline tests. Diagnosis in late pregnancy means a high-risk infant
When did you start ART, which regimen, any changes? TLD, TLE400, ABC-based?	Less than 4 weeks of ART before delivery = high-risk infant. Past regimen changes suggest past treatment failure
Last viral load: result AND date?	The most important question. Decides infant prophylaxis and whether IAC is needed. A suppressed VL from 9 months ago is not reassuring
Adherence: doses missed in the last month (use a calendar), who reminds you, where you keep your drugs, vomiting (hyperemesis), stock-outs, alcohol, depression	Adherence is the main cause of non-suppression. Probe non-judgementally

Question	Why it matters
Disclosure, partner status, safety: has your partner tested? on ART or PrEP? any violence?	Partner testing, discordance (partner may need PrEP), social support, safety (intimate partner violence), adherence
TB symptom screen: cough, fever, night sweats, poor weight gain	At **every** visit. Any one positive = investigate; TB drugs change the DTG dose
Neurological: headache, confusion, neck stiffness, fits	Cryptococcal or TB meningitis: **defer** ART start (section 9)
Rash, jaundice, vomiting, abdominal pain	Drug toxicity; hepatitis B flare
Other medicines: rifampicin, anticonvulsants, iron/calcium/antacids, metformin	**Drug interactions with DTG** cause non-suppression
On co-trimoxazole?	If yes, **no IPTp-SP**; give an ITN
Triple elimination results: syphilis (treated? how many doses?), HBsAg	Decides the penicillin course, TDF-based ART, and the HBV birth dose
Gestational age and dating: LNMP, early scan	Decides when the pre-delivery VL is due (around 34–36 weeks)
Previous children: HIV status, EID results, feeding	A previously infected child suggests previous poor control; shows how well she knows the programme
Birth and feeding plan: where she will deliver, exclusive breastfeeding	Ensures prophylaxis is ready and feeding is safe

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination. A sick or labouring woman gets the ABCDE assessment first (section 9).

Introduction, consent and preparation

- Introduce yourself, confirm her identity and gestation, explain the examination and gain consent; offer a chaperone for abdominal, breast and genital examination. Respect confidentiality: do not discuss her status in front of relatives unless she has disclosed to them.
- **Hand hygiene** (and standard precautions; gloves for genital examination).
- **Position and exposure**: sitting, then semi-recumbent with **left lateral tilt** for the abdomen; expose only what you are examining.

General inspection and vital signs

Sign	Mechanism	What it means
Well or unwell; breathlessness	Pneumonia, PCP, TB, anaemia	Sick woman: ABCDE first
Wasting, poor weight gain (track the trend)	Chronic infection, TB, poor intake	TB until proven otherwise; WHO stage 3–4
Pallor	HIV, malaria, previous AZT, iron deficiency	Anaemia: Hb, iron, plan for PPH
Jaundice	Drug-induced hepatitis, HBV flare, malaria	LFTs, HBsAg
Skin: **Kaposi sarcoma**, herpes zoster scars, papular pruritic eruption	Immunosuppression	WHO stage 2–4
Confusion, drowsiness	Meningitis (cryptococcal, TB), cerebral malaria, sepsis	Urgent CrAg, LP, mRDT

Vital signs: pulse, **BP** (HIV-positive women also get pre-eclampsia), RR, **temperature** (fever: malaria first), **SpO_2** if breathless (a clear chest with low SpO_2 suggests PCP); weight and BMI.

Hands, face and neck

- Hands: clubbing (chronic lung disease, TB), palmar pallor, nail candida.
- Face and mouth: conjunctival pallor, scleral jaundice, **oral thrush**, **oral hairy leukoplakia**, oral ulcers, parotid enlargement.
- Neck: **lymphadenopathy** (persistent generalised lymphadenopathy versus large, matted or tender TB nodes), **neck stiffness** (meningitis).

Chest

Crackles, bronchial breathing, effusion (TB, pneumonia); clear chest with low SpO_2 suggests PCP.

Obstetric abdominal examination

- **Inspection**: size for dates, scars; liver or spleen visible.
- **Palpation**: liver tenderness (hepatitis), uterine tenderness (chorioamnionitis).
- **Fundal height (SFH) versus dates**: **growth restriction is more common**.
- **Lie and presentation**.
- **Engagement** (fifths palpable).
- **Auscultation of the fetal heart** for a full minute (fetal tachycardia with maternal fever suggests chorioamnionitis).

Neurological

Neck stiffness, focal deficits (meningitis, toxoplasmosis), GCS.

Completing the examination

- **Genital (speculum where indicated)**: ulcers (syphilis, herpes), warts, discharge. Genital infections raise genital VL and transmission risk.
- **In labour**: cervical dilatation, membranes intact or ruptured and **for how long**, liquor colour. Time of ROM matters: plan to shorten the interval to delivery.
- **Bedside tests**: urine dipstick, Hb, mRDT if febrile, rapid tests (section 8).
- Thank her, hand hygiene, summarise.

Document (positives and key negatives)

Area	Model documentation
General	"Not pale, no jaundice, no oral thrush, no lymphadenopathy, weight 64 kg (up 2 kg in 4 weeks)"
Chest	"Chest clear, SpO_2 98% on air"
Neurological	"No neck stiffness, GCS 15"
Obstetric	"SFH 34 cm at 35 weeks, cephalic, FHR 140"
Genital	"No ulcers, no abnormal discharge"

Red flags

RED FLAG

- **Cough for 2 weeks or more, fever, night sweats, failure to gain weight**: TB until proven otherwise; do GeneXpert.
- **Headache, fever, confusion, neck stiffness**: cryptococcal or TB meningitis; do CrAg, LP; defer ART start by 4–6 weeks for cryptococcal meningitis.
- **Breathlessness with dry cough and desaturation** (SpO_2 below 94%, RR above 30): PCP, pneumonia or TB.
- **Jaundice, vomiting, right upper quadrant pain** after starting ART: drug-induced hepatitis or HBV flare.
- **Fever ≥38.0°C**: malaria first, then pyelonephritis, chorioamnionitis, pneumonia, TB.
- **VL of 1,000 copies/mL or more in the third trimester**: high-risk infant; urgent adherence intervention.
- **Newly diagnosed in labour or postpartum**: the baby is high-risk; start mother on TLD and baby on AZT + NVP now, not tomorrow.
- **Advanced HIV disease** (CD4 below 200): needs CrAg screening, TB LAM, and close follow-up.

Thinking like a clinician

- **"Positive test in pregnancy" is not the end of the history; it is the start.** Every HIV-positive pregnant woman on the ward should be summarised with four facts: **regimen, duration on ART, last VL (result and date), and adherence.** Then decide infant risk.
- **If VL is unknown near term, treat as high-risk.** Do not delay the baby's prophylaxis waiting for a result.
- **If the woman has been on ART for years and is now non-suppressed, think adherence first**, then drug interactions (for example, rifampicin, or iron taken with DTG), then resistance. On a DTG regimen, resistance is uncommon.
- **Negative at booking does not mean negative at delivery.** Seroconversion in pregnancy carries a very high transmission risk because acute HIV produces enormous viral loads. That is why we **retest**.
- **Fever in an HIV-positive pregnant woman in Uganda**: malaria, UTI/pyelonephritis, pneumonia, TB, chorioamnionitis, puerperal sepsis, meningitis. Always do an mRDT/blood smear first.
- **Weight loss is hidden by pregnancy**: a woman who is "not gaining" may be losing; plot the weight trend.
- **Common student mistakes**:
  1. Forgetting that **co-trimoxazole and IPTp-SP must not be combined**.
  2. Recommending caesarean section "because she is HIV-positive". Not in Uganda unless there is an obstetric indication.
  3. Forgetting to ask the **date** of the last VL (a suppressed VL 9 months ago is not reassuring).
  4. Advising formula feeding. Uganda recommends breastfeeding with ART.
  5. Saying the infant needs an antibody test at 6 weeks. Antibodies are maternal until about 18 months; use **DNA PCR**.

CLINICAL PEARL

Before you present an HIV-positive woman, say the **four facts** (regimen, time on ART, last VL with date, adherence) and then say the **infant risk category out loud**. That is exactly what the consultant will ask for.

7. Differential Diagnoses

This topic is about managing a known diagnosis, so the useful differentials are **the causes of a sick HIV-positive pregnant woman** and **the causes of a non-suppressed viral load**.

The unwell HIV-positive pregnant woman

Differential	Distinguishing clinical feature	Distinguishing investigation
Malaria	Fever, rigors, headache; any trimester	mRDT or blood smear
Pulmonary TB	Cough 2 weeks or more, night sweats, poor weight gain, haemoptysis	GeneXpert on sputum; TB LAM if CD4 below 200 or seriously ill; CXR with abdominal shielding
PCP	Subacute dry cough, breathlessness, desaturation on exertion, often not on CTX	SpO_2, CXR (bilateral interstitial), clinical response to high-dose co-trimoxazole
Bacterial pneumonia	Acute fever, productive cough, focal crackles	CXR, FBC
Cryptococcal meningitis	Headache, fever, reduced consciousness; CD4 below 100	Serum CrAg, LP with CSF CrAg, opening pressure
Pyelonephritis	Fever, loin pain, dysuria	Urinalysis, urine culture
Chorioamnionitis / puerperal sepsis	Fever, uterine tenderness, offensive liquor/lochia, fetal tachycardia	FBC, blood culture, HVS
Drug-induced hepatitis / HBV flare	Jaundice, RUQ pain after ART start	LFTs, HBsAg

Why is the viral load not suppressed?

Cause	Clues	Action
Poor adherence (most common)	Missed doses, pill counts off, stigma, non-disclosure, hyperemesis	IAC, treatment supporter, address barriers
Recently started ART	Started less than 3 months ago	Recheck at 3 months after start
Drug interaction	Rifampicin, carbamazepine/phenytoin, iron/calcium/antacids taken with DTG	Adjust DTG dose or timing
Vomiting / malabsorption	Hyperemesis, diarrhoea	Antiemetics, treat cause
Drug resistance	Persistent non-suppression despite verified adherence	Discuss with ART clinic/switch committee; resistance testing where available

8. Clinical Workup

The workup of a woman with HIV in pregnancy has two jobs: **keep the mother well** (find the opportunistic infection, the drug problem, the non-suppression) and **classify the baby's risk** before delivery. Most of this happens in the eMTCT clinic, but the same steps apply when she is admitted to the antenatal ward or arrives in labour. "Available at" reflects the source notes Verify *availability compiled from the source notes, not an MoH list*.

Bedside

Test (minutes, on the ward or in the clinic)	Why / expected finding	Available at
HIV rapid tests (national serial algorithm: screening test, then confirmatory test, then tie-breaker; the kits change, so follow the current algorithm)	Diagnosis. A single reactive screening test is **not** a diagnosis. Always confirm a first positive; retest before ART start per programme rules to avoid misdiagnosis	HC II upwards

Test (minutes, on the ward or in the clinic)	Why / expected finding	Available at
Dual HIV/syphilis RDT	Syphilis screen. The treponemal test **stays positive after treatment**, so ask about past treatment before re-treating	HC II/III upwards
HBsAg RDT	Chronic HBV: TDF-containing regimen, HBV birth dose within 24 hours	HC III upwards where stocked
Point-of-care CD4 (semi-quantitative, available at many sites)	Above or below 200 cells/mm³: identifies advanced HIV disease the same day	Many HC IVs and hospitals
Serum CrAg lateral-flow test	If CD4 is 200 or below: positive without meningitis = pre-emptive fluconazole (senior advice in the first trimester)	HC IV and hospitals
Urine TB LAM	CD4 below 200 or seriously ill: a positive result allows TB treatment the same day	HC IV and hospitals
mRDT	Every fever. Co-infection increases placental inflammation and transmission	All levels
HemoCue / Hb	Anaemia (HIV, malaria, previous AZT)	HC III upwards
Urine dipstick	Protein (pre-eclampsia, TDF kidney effect), nitrites/leucocytes (UTI), glucose	All levels
RBS	If symptoms or risk factors; DTG may modestly raise glucose; interaction with metformin	HC III upwards
Fetal heart / CTG / partograph	Fetal wellbeing; in labour, plot on the partograph and avoid prolonged labour	All levels (CTG at hospitals)

Basic lab

Test	When	Expected / abnormal finding	How it changes management (and where)
Viral load	First ANC visit if already on ART; **3 months after starting ART**; then **every 3 months** until breastfeeding ends; make sure one is available at **34–36 weeks**	Below 50 = undetectable; 50–999 = low-level viraemia; **1,000 or more = non-suppressed**	Non-suppressed: IAC ×3, check interactions, repeat VL; **baby high-risk** if not suppressed near delivery. POC VL at many HC IVs and hospitals; otherwise DBS/plasma via the hub to CPHL (results can take weeks, so **order early**)
CD4 count	Baseline at diagnosis; if unwell or re-engaging after defaulting	Below 200 = advanced HIV disease	CrAg, TB LAM, closer follow-up; CTX is already given to all pregnant women. Full CD4 counts mainly at hospitals; POC CD4 at many HC IVs
FBC	Booking, then per ANC schedule; any fever	Anaemia; neutropenia/anaemia if on AZT; raised WBC in sepsis	Iron, transfusion planning, change AZT. HC IV upwards
Creatinine / eGFR	Baseline where available, especially before TDF; on TDF-based PrEP	CrCl below 50 mL/min	Switch TDF to ABC-based regimen (**check HBsAg first**). Regional/national referral mostly

Test	When	Expected / abnormal finding	How it changes management (and where)
LFTs	Jaundice, vomiting, RUQ pain, HBsAg-positive	Raised ALT/AST, bilirubin	Drug-induced hepatitis versus HBV flare; stop the offending drug only with senior advice (do not stop TDF abruptly in HBV). Hospital level
Blood group and Rh	Booking	Rh-negative	Anti-D plan. HC IV upwards
Urine culture	Dipstick positive or symptoms	Significant growth	Treat UTI/pyelonephritis. Hospital level
Blood glucose	Where indicated	Hyperglycaemia	DTG can modestly raise glucose; interaction with metformin
Infant DNA PCR	4–6 weeks, 9 months, 6 weeks after breastfeeding stops (± at birth in high-risk infants where available)	Positive	Start ART the same day and send a confirmatory sample. Antibody tests are unreliable before 18 months because of maternal IgG. DBS from HC III upwards; POC EID at many HC IVs and hospitals

UGANDA CONTEXT

The **1,000 copies/mL** cut-off is the long-standing Uganda programme threshold for "non-suppressed", and it is what the rest of these notes use. Newer WHO guidance puts more emphasis on reaching **below 50** (undetectable) and treats **50–999** as low-level viraemia that still needs adherence support; some units act at **200**. Check the current Uganda Consolidated Guidelines and your clinic's rule, and flag any detectable VL near term to a senior.

Imaging

Test	When	Expected finding	How it changes management (and where)
Chest X-ray (with abdominal shielding)	Respiratory symptoms, suspected TB/PCP	Cavities, effusion, miliary pattern (TB); bilateral interstitial shadows (PCP)	Pregnancy is **not** a reason to withhold a needed CXR. Hospital level
Obstetric ultrasound	Dating, growth concerns	FGR, oligohydramnios	Fetal surveillance, timing of delivery. HC IV and hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management (and where)
Serum CrAg (± LP with CSF CrAg)	CD4 200 or below; any headache/confusion	Positive	Pre-emptive fluconazole, or LP and induction therapy if meningitis. **Delay ART 4–6 weeks** in cryptococcal meningitis. LP at hospital level
GeneXpert (sputum)	Any positive TB symptom screen	MTB detected ± rifampicin resistance	Start TB treatment; **DTG 50 mg twice daily** while on rifampicin and for 2 weeks after; TPT only once TB is excluded. HC IVs and hospitals via the hub

Test	When	Expected / abnormal finding	How it changes management (and where)
HBV DNA / HBeAg	HBsAg-positive (where available)	HBV DNA 200,000 IU/mL or more, or HBeAg-positive	TDF from about 28 weeks is a priority; where unavailable, many programmes offer TDF to all HBsAg-positive women. Rarely available outside national referral level
HIV drug-resistance test	Persistent non-suppression despite verified good adherence	Resistance mutations	Via the **switch committee**; sample to CPHL

Putting it together

Model one-line summary

> "A 24-year-old G2P1 at 35 weeks, HIV-positive diagnosed at booking at 20 weeks, on TLD for 15 weeks, with a VL of 2,400 copies/mL at 33 weeks, self-reports missing 'some' doses, co-trimoxazole prophylaxis, HBsAg and syphilis negative, TB screen negative, clinically well."

Problem list

1. HIV in pregnancy, **non-suppressed** VL in the third trimester.
2. **High-risk HIV-exposed infant** anticipated.
3. Adherence barriers to explore (disclosure, stigma, iron timing with DTG).
4. Routine ANC at 35 weeks (birth plan, feeding plan).

Working diagnosis: HIV infection on first-line TLD with non-suppression, most likely from poor adherence (recent start, missed doses) rather than resistance.

Model case presentation (ward round)

> "Mrs N is a 26-year-old G3P2+0 at 36 weeks by dates, known HIV-positive since 2019, on **TLD for 5 years**, with a **last VL of below 50 copies/mL in June** (3 months ago), good self-reported adherence and no missed doses. She is on **co-trimoxazole 960 mg daily** and therefore not on IPTp. HBsAg and syphilis negative this pregnancy. TB screen negative. She has disclosed to her husband, who is HIV-negative and on PrEP. Obstetrically she is well, SFH corresponds to dates, cephalic, fetal heart normal. **Plan**: repeat VL now so it is available at delivery; facility delivery; if VL remains suppressed the baby is **low-risk** and will receive **NVP 15 mg once daily for 6 weeks**, first PCR at 6 weeks; exclusive breastfeeding."

Worked example 1: reading a VL/CD4/HBsAg panel (the 24-year-old above)

Result	What it means
VL at 33 weeks: **2,400 copies/mL**	Non-suppressed (1,000 or more). Only 13 weeks on ART, so partly "not yet suppressed", but missed doses too
Baseline CD4: **310 cells/mm³**	Not advanced HIV disease; CrAg not required by the CD4 rule
HBsAg: **negative**	No HBV action; routine infant immunisation
Syphilis: **negative**	Retest is not routine, but treat if there is a new genital ulcer

Plan:

1. Start **intensive adherence counselling** today (session 1), with a treatment supporter; check that ferrous sulphate is taken **2 hours after or 6 hours before** DTG, or together with food.
2. Look for other interactions (rifampicin, anticonvulsants): none.

3. **Repeat VL** early so a result is available before delivery (the clinic may repeat within about 4 weeks late in pregnancy rather than waiting the full 3 months).
4. **Classify the baby as high-risk** unless a VL below 1,000 copies/mL is available from the last 4 weeks before delivery: **AZT + NVP for 6 weeks, then NVP alone for 6 weeks**; if she weighs 2.5 kg or more, both drugs at **15 mg (1.5 mL)**.
5. Consider a birth PCR for the baby where available; first routine PCR at 4–6 weeks.
6. Facility delivery; avoid prolonged ROM, routine ARM and episiotomy. Mode of delivery by obstetric indication.

Worked example 2: new diagnosis in labour

A 19-year-old primigravida at 38 weeks, 6 cm dilated, **no ANC**. Rapid HIV test: screening test reactive, confirmatory test reactive. POC CD4 **below 200**. HBsAg **reactive**. Syphilis RDT **reactive**.

Finding	Interpretation	Action
HIV positive, no ART	Untreated, high VL likely; ART under 4 weeks before delivery	**TLD now**; baby **high-risk**: AZT + NVP within 6 hours of birth, 12 weeks in total
CD4 below 200	Advanced HIV disease	Serum CrAg and urine TB LAM; TB symptom screen; co-trimoxazole 960 mg daily; ask about headache
HBsAg positive	HIV/HBV co-infection	TLD already contains TDF + 3TC, which treat HBV. **Never use an ABC-based regimen alone**. Baby: **HBV birth-dose vaccine within 24 hours** (± HBIG where available)
Syphilis RDT positive, never treated	Untreated syphilis, duration unknown	**Benzathine penicillin 2.4 MIU IM weekly × 3** (start now). Examine the baby for congenital syphilis and treat the newborn per the paediatric protocol, because treatment less than 30 days before delivery does not protect the fetus

9. Management

Immediate / emergency

First 5 minutes (sick or labouring woman)

1. **Is she sick or in labour?** Check **pulse, BP, RR, temperature, SpO_2**, level of consciousness, and whether she is contracting or bleeding. Work out the shock index (HR ÷ SBP; ≥0.9 is abnormal) if she is bleeding.
2. **Danger signs that need a senior now** (ABCDE):
   - **A/B**: **breathless with SpO_2 below 94%**, or RR above 30: PCP, pneumonia or TB. Oxygen, sit her up, call for help.
   - **C**: bleeding or shock: shock index, IV access, resuscitate as for any obstetric emergency.
   - **D**: **headache with fever, confusion or neck stiffness**: cryptococcal or TB meningitis. Do not start ART today; arrange CrAg and LP.
   - **E**: **fever ≥38.0°C**: mRDT or blood smear first, then look for pyelonephritis, chorioamnionitis, pneumonia, TB. **Jaundice, vomiting, RUQ pain**: drug-induced hepatitis or an HBV flare.
3. **In labour?** Ask straight away: **"Are you on ART? Did you take today's dose? What was your last viral load, and when?"** If her status is **unknown**, do the rapid HIV test now; do not wait for the ward round.
4. **Start at once**: if newly positive in labour, give **TLD now** and make sure **AZT + NVP syrup** is on the ward for the baby. If she is known positive, make sure she has **taken today's TLD** (even on the day of CS) and the infant syrups are ready.

Staff exposure: standard precautions for staff; post-exposure prophylaxis (**TLD for 28 days**) after a needlestick.

Definitive

1. Testing in ANC (triple elimination)

1. **Offer HIV/syphilis dual test and HBsAg at the first ANC visit** (provider-initiated, opt-out, with pre-test information and post-test counselling). Test the **partner** too (couple testing, assisted partner notification).
2. **If HIV-negative**: counsel on prevention; **retest in the third trimester** (or in labour/immediately postpartum if missed), and continue periodic retesting during breastfeeding (at the 6-week postnatal visit and then according to the national schedule). Offer **PrEP** if at substantial risk.
3. **If HIV-positive**: confirm with the national algorithm, then **start ART the same day** after counselling (see below).
4. **If syphilis-positive**: treat on the spot (subsection 5) and treat the partner.
5. **If HBsAg-positive**: link to hepatitis care; decide on TDF (subsection 5); plan the infant's birth-dose vaccine within 24 hours.

> UGANDA CONTEXT
> **Retesting HIV-negative women is part of eMTCT.** Women who seroconvert in pregnancy or breastfeeding now account for a large proportion of infant infections in Uganda. A booking test is not enough; retest in the third trimester, in labour if missed, and during breastfeeding.

2. Starting ART (all settings, HC III upwards)

1. **Counsel** (diagnosis, lifelong treatment, benefits for her and the baby, disclosure, partner testing, adherence). Identify a **treatment supporter**.
2. **Baseline tests**: CD4, HBsAg, syphilis, Hb, creatinine if available, TB screen, CrAg if CD4 200 or below. **Do not delay ART for results** except where meningitis is suspected.
3. **Start TLD the same day** (rapid ART initiation), or within 7 days at most.
4. **Screen for TB** before starting TB preventive therapy; if symptomatic, investigate first.
5. **Give co-trimoxazole prophylaxis** per national guidance (all HIV-positive pregnant and breastfeeding women in Uganda are eligible) and **do not give IPTp-SP**.
6. **Continue routine ANC**: ferrous sulphate + folic acid (timed away from DTG), tetanus-diphtheria vaccine, mebendazole after the first trimester, ITN.
7. **Schedule the first VL at 3 months after starting ART.**

ART regimens

Situation	Regimen	Dose	Notes
Preferred first-line (all, including pregnant and breastfeeding women)	**TLD**: tenofovir DF + lamivudine + dolutegravir	TDF 300 mg / 3TC 300 mg / DTG 50 mg, **one tablet orally once daily**	Rapid suppression, high resistance barrier; treats HBV too
DTG not suitable (intolerance, unmanageable interaction)	**TLE400**: TDF + 3TC + efavirenz 400 mg	One tablet once daily, at night	EFV causes vivid dreams/dizziness; **reduces implant efficacy**; lower resistance barrier
TDF contraindicated (CrCl below 50 mL/min)	**ABC + 3TC + DTG**	ABC 600 mg / 3TC 300 mg + DTG 50 mg once daily	ABC does **not** treat HBV; check HBsAg
On rifampicin-based TB treatment	TLD **plus an extra DTG 50 mg**	DTG 50 mg **twice daily**	Continue twice daily until **2 weeks after** stopping rifampicin
Already on another suppressed regimen when she conceives	**Continue**		Transition to TLD per programme; do not interrupt

DRUG DOSING

DTG interactions you must know

- **Rifampicin** induces DTG metabolism: give **DTG 50 mg twice daily** during TB treatment and for 2 weeks after.
- **Polyvalent cations** (iron, calcium, magnesium/aluminium antacids, zinc) chelate DTG in the gut: take DTG **2 hours before or 6 hours after** them, or take them together **with food**. Ferrous sulphate/folic acid tablets in ANC are the everyday example.
- **Metformin**: DTG raises metformin levels; use the lowest effective metformin dose and monitor.
- **Carbamazepine, phenytoin, phenobarbital**: reduce DTG levels; consult.

When to delay ART

- **Cryptococcal meningitis**: delay ART **4–6 weeks** after starting antifungal treatment (early ART raises mortality from immune reconstitution).
- **TB meningitis**: delay by about 4–8 weeks per specialist advice.
- **Pulmonary TB**: start TB treatment first, then ART **within 2 weeks** (sooner if CD4 is very low).

3. Antenatal monitoring

When	What
Every visit	Adherence check (missed doses, pill count), TB symptom screen, side effects, weight, BP, SFH, fetal heart, danger signs, CTX refill, disclosure/partner status
Viral load	**Already on ART at conception**: VL at the **first ANC visit**. **Started in pregnancy**: VL **3 months after starting**. Then **every 3 months until breastfeeding ends**
Around **34–36 weeks**	Make sure a recent VL is available to classify the baby's risk; if the last VL is non-suppressed or unknown, plan high-risk infant prophylaxis
Birth plan	Facility delivery; NVP (and AZT) syrup ready; infant feeding counselling

What to do with a non-suppressed VL (1,000 copies/mL or more)

1. **Do not switch immediately.** Most non-suppression is adherence.
2. Start **intensive adherence counselling (IAC)**: at least 3 sessions, about a month apart. Explore barriers (stigma, non-disclosure, alcohol, depression, vomiting, pill burden, food, distance, intimate partner violence), set up a treatment supporter, directly observed doses if needed.
3. Check for **interactions** (rifampicin, iron timing, anticonvulsants).
4. **Repeat VL** after completing IAC (about 3 months after the first result; in late pregnancy the clinic may repeat sooner so the result is available before delivery).
5. If **still non-suppressed** despite good adherence: present to the **ART switch committee** (hospital level) for regimen change, with resistance testing where available.
6. Regardless of the repeat, **classify the baby as high-risk** if the VL near delivery is not suppressed.

EXAM FAVOURITE

VL thresholds: **below 50** = undetectable; **50 to 999** = low-level viraemia (suppressed by programme definition but needs adherence support); **1,000 or more** = non-suppressed (IAC then repeat).

4. Opportunistic infection prevention

Drug	Who	Dose	Notes
Co-trimoxazole	All HIV-positive pregnant and breastfeeding women (Uganda)	**960 mg orally once daily**	Protects against PCP, toxoplasmosis, malaria and bacterial infection. **Replaces IPTp-SP**
Isoniazid preventive therapy (6H)	HIV-positive, TB excluded	**Isoniazid 300 mg orally once daily for 6 months + pyridoxine 25 mg daily**	Preferred TPT in pregnancy; 3HP (isoniazid + rifapentine weekly for 12 weeks) has limited pregnancy data
Fluconazole (pre-emptive)	Positive serum CrAg without meningitis	Per national AHD guideline	High-dose fluconazole in the **first trimester is teratogenic**; seek senior advice

RED FLAG

Never give IPTp-SP to a woman on co-trimoxazole. Both are sulfonamide/antifolate drugs: combining them adds no benefit and raises the risk of severe skin reactions (Stevens-Johnson syndrome) and neonatal kernicterus. Co-trimoxazole itself protects against malaria. Give her an ITN.

5. Syphilis and hepatitis B (triple elimination)

Condition	Treatment	Dose	Notes
Syphilis (early, under 2 years)	**Benzathine penicillin G**	**2.4 million IU IM** (1.2 MIU in each buttock) **single dose**	Treat the partner. Treatment at least 30 days before delivery prevents most congenital syphilis
Syphilis, late or unknown duration	Benzathine penicillin G	2.4 MIU IM **once weekly for 3 weeks**	Most ANC cases are of unknown duration
Penicillin allergy	Erythromycin	500 mg orally 6-hourly for 14 days	Does not reliably treat the fetus: **treat the baby after birth**
HBsAg-positive and HIV-positive	TLD (TDF + 3TC treat HBV)	As above	Never use a regimen without TDF (e.g. ABC-based) without specialist input; stopping TDF can cause HBV flare
HBsAg-positive, HIV-negative	Tenofovir DF (prophylaxis to cut MTCT)	**300 mg orally once daily from about 28 weeks**, continued at least until delivery (often to 12 weeks postpartum)	Priority if HBV DNA is 200,000 IU/mL or more or HBeAg-positive; where these tests are unavailable, national/WHO guidance increasingly supports offering TDF to all HBsAg-positive pregnant women. Assess her own need for lifelong treatment
Infant of HBsAg-positive mother	Hepatitis B birth-dose vaccine (**within 24 hours**) ± HBIG where available		Then the routine pentavalent series

6. PrEP in pregnancy and breastfeeding

- **Who**: HIV-negative pregnant or breastfeeding women at substantial risk: HIV-positive partner who is not suppressed or of unknown status, multiple partners, sex work, recent STI, partner at high risk (e.g. fishing communities, truck drivers).

- **What**: oral **TDF/FTC (or TDF/3TC) one tablet once daily**. Safe in pregnancy and breastfeeding. Protection takes about 7 days of daily dosing for vaginal exposure.
- **Monitoring**: HIV test before starting and **every 3 months**; creatinine where available; adherence; check HBsAg.
- If she seroconverts on PrEP, stop PrEP and start full ART (TLD) immediately.

7. Intrapartum care

1. **Continue ART in labour**: she takes her usual TLD dose, even on the day of caesarean section.
2. **Unknown status in labour**: offer a rapid HIV test; if positive, **start TLD immediately** and classify the baby as high-risk.
3. **Use the partograph**, avoid prolonged labour.
4. **Avoid**:
   - **Prolonged rupture of membranes** (risk rises with each hour after about 4 hours);
   - **Routine artificial rupture of membranes** (do it only for a clear obstetric indication);
   - **Routine episiotomy**;
   - **Fetal scalp electrodes / scalp blood sampling**;
   - **Instrumental delivery** unless clearly indicated (if needed, ventouse or forceps may be used; weigh against the risk of CS);
   - **Unnecessary vaginal examinations**.
5. **Mode of delivery**: **vaginal delivery** is appropriate for almost all women; **caesarean section for obstetric indications**. In well-resourced settings, planned CS is offered when VL is high near term; in Uganda this is not routine policy and CS carries its own risks (sepsis, PPH, future uterine rupture). Discuss with the consultant if VL is very high at term.
6. **Active management of the third stage** as usual (oxytocin 10 IU IM).
7. **Newborn**: wipe (do not scrub) the baby; clamp the cord after 1–3 minutes as usual; suction only if needed; skin-to-skin; **start breastfeeding within 1 hour**; give **prophylaxis within 6 hours** (as soon as possible; benefit falls after 72 hours, but a late-presenting baby should still be started); vitamin K, eye prophylaxis, BCG and OPV-0; HBV birth dose.

> **NOTE**
> Rupture of membranes before labour at term in an HIV-positive woman: **expedite delivery** (induce with oxytocin if favourable) rather than waiting, to shorten the exposure time.

8. Infant prophylaxis

Risk category: section 5 (high-risk = late diagnosis, ART under 4 weeks, VL 1,000 or more or unknown in the last 4 weeks, seroconversion).



Risk	Regimen	Duration
Low-risk	**NVP** once daily	**6 weeks**
High-risk	**AZT twice daily + NVP once daily** for **6 weeks**, then **NVP alone** for a further **6 weeks**	**12 weeks** in total. If the mother's VL at around 12 weeks postpartum is still 1,000 copies/mL or more, **continue NVP** until she is suppressed (or until 4 weeks after breastfeeding stops)
Mother diagnosed while breastfeeding	Start mother on TLD, give baby **AZT + NVP**, do a **DNA PCR now**; treat as high-risk	12 weeks (as above)

Doses (WHO weight bands, as used in Uganda)

Age / weight	NVP (10 mg/mL syrup), once daily	AZT (10 mg/mL syrup), twice daily
Birth to 6 weeks, **2.0–2.49 kg**	**10 mg (1 mL)**	**10 mg (1 mL)**
Birth to 6 weeks, **2.5 kg or more**	**15 mg (1.5 mL)**	**15 mg (1.5 mL)**
6 to 12 weeks	**20 mg (2 mL)**	Not given (AZT stops at 6 weeks)
Extended NVP only (mother still non-suppressed): **12 weeks to 6 months** / **6–9 months** / **9 months to end of breastfeeding**	**25 mg** / **30 mg** / **40 mg** (WHO age bands; confirm with the national paediatric chart) Verify *extended NVP age-band doses against the current Uganda paediatric dosing chart*	Not given
Below 2.0 kg	mg/kg dosing per national paediatric chart; discuss with the paediatrician	

DRUG DOSING

Start infant prophylaxis **as soon as possible after birth, ideally within 6 hours**. A baby born at night must not wait for the morning round. Show the mother how to measure the syrup with the syringe and give her enough to last until the next visit.

9. Infant follow-up: EID, co-trimoxazole, feeding

Early infant diagnosis (Uganda schedule)

When	Test
4–6 weeks (first immunisation visit)	**1st DNA PCR** (DBS or POC). High-risk infants may also have a PCR at birth where available
9 months	**2nd DNA PCR**
6 weeks after complete cessation of breastfeeding	**3rd DNA PCR**
18 months (18–24 months)	**Rapid antibody test**
Any time the child is unwell with signs suggestive of HIV	DNA PCR

- **Any positive PCR**: start ART the same day, collect a **confirmatory DBS** at the same time.
- **Final status** is declared only after the last test done at least 6 weeks after all breastfeeding stops.

Co-trimoxazole for the HEI

- Start at **6 weeks** of age and continue until HIV infection is excluded after breastfeeding ends.
- Dose: **120 mg once daily** under 6 months; **240 mg once daily** from 6 months to 5 years.

Infant feeding

1. **Exclusive breastfeeding for the first 6 months**: no water, glucose water, herbs or formula.
2. **At 6 months** add appropriate complementary foods and **continue breastfeeding to at least 12 months** (and up to 24 months), provided the mother is on ART and adherent.
3. **Stop breastfeeding gradually** over about a month when she stops; the final PCR is 6 weeks later.
4. Replacement (formula) feeding only if it is **AFASS** (acceptable, feasible, affordable, sustainable and safe), which is rare in Uganda.
5. Treat **mastitis, cracked nipples and infant thrush** promptly (they raise breast-milk transmission).

> **UGANDA CONTEXT**
> Unlike the UK (where formula feeding is advised for HIV-positive mothers), **Uganda and WHO recommend breastfeeding**, because in our setting the risks of diarrhoea, pneumonia and malnutrition from formula far outweigh the small transmission risk from a suppressed mother.

10. Postpartum and long term

- Continue **TLD for life**; continue VL every 3 months while breastfeeding, then per adult schedule (every 6–12 months once stable).
- **Family planning**: DTG has **no interaction** with hormonal contraceptives. EFV reduces implant (and oral contraceptive) levels; advise an IUD, DMPA or condoms in addition.
- **Cervical cancer screening** (VIA or HPV test) as for all women with HIV.
- **Differentiated service delivery**: stable mothers can join community drug distribution points or fast-track refills once suppressed.
- Keep the **mother-baby pair** linked until the child's final status is known.

Drug table

Drug	Dose (adult; infant per weight band)	Route	Frequency	Duration	Key cautions
TLD (TDF 300 mg / 3TC 300 mg / DTG 50 mg)	One tablet	Oral	Once daily	Lifelong; start same day	Separate from iron/calcium/antacids (2 h before or 6 h after, or with food); TDF renal (CrCl below 50: switch to ABC); stopping TDF can flare HBV
Extra DTG (with rifampicin)	50 mg (total DTG 50 mg twice daily)	Oral	12 h after the TLD dose	Until 2 weeks after rifampicin stops	Rifampicin induces DTG metabolism
TLE400 (TDF + 3TC + EFV 400 mg)	One tablet	Oral	Once daily, at night	Lifelong	Vivid dreams, dizziness; reduces implant and oral contraceptive efficacy
ABC + 3TC + DTG	ABC 600 mg / 3TC 300 mg + DTG 50 mg	Oral	Once daily	Lifelong	Does not treat HBV; check HBsAg

Drug	Dose (adult; infant per weight band)	Route	Frequency	Duration	Key cautions
Co-trimoxazole (mother)	960 mg	Oral	Once daily	Through pregnancy and breastfeeding (per national guidance)	**No IPTp-SP**; rash (SJS)
Isoniazid + pyridoxine (6H)	Isoniazid 300 mg + pyridoxine 25 mg	Oral	Once daily	6 months	TB excluded first; hepatitis
Fluconazole (pre-emptive, CrAg-positive)	Per national AHD guideline	Oral	Daily	Per guideline	High dose teratogenic in the first trimester: senior advice
Benzathine penicillin G	2.4 MIU (1.2 MIU each buttock)	IM	Single (early syphilis) or weekly	1 dose, or 3 weeks (late/unknown)	Treat the partner; under 30 days before delivery does not protect the fetus
Erythromycin (penicillin allergy)	500 mg	Oral	6-hourly	14 days	Does not treat the fetus: treat the baby
Tenofovir DF (HBsAg-positive, HIV-negative)	300 mg	Oral	Once daily	From about 28 weeks to delivery (often 12 weeks postpartum)	Assess her own need for lifelong treatment
TDF/FTC or TDF/3TC (PrEP)	One tablet	Oral	Once daily	While at risk	HIV test every 3 months; about 7 days to protect; creatinine, HBsAg
TLD (PEP)	One tablet	Oral	Once daily	28 days	Start as soon as possible after exposure
Oxytocin	10 IU	IM	Once (third stage)		Active management as usual
Nevirapine syrup (10 mg/mL), infant	2.0–2.49 kg: 10 mg (1 mL); 2.5 kg or more: 15 mg (1.5 mL); 6–12 weeks: 20 mg (2 mL); extended: 25 / 30 / 40 mg by age band	Oral	Once daily	6 weeks (low-risk); 12 weeks or longer (high-risk)	Start within 6 hours of birth; teach syringe measurement
Zidovudine syrup (10 mg/mL), infant	2.0–2.49 kg: 10 mg (1 mL); 2.5 kg or more: 15 mg (1.5 mL)	Oral	Twice daily	6 weeks (high-risk only)	Anaemia, neutropenia
Co-trimoxazole (infant)	Under 6 months: 120 mg; 6 months to 5 years: 240 mg	Oral	Once daily	From 6 weeks until HIV excluded after weaning	

Drug	Dose (adult; infant per weight band)	Route	Frequency	Duration	Key cautions
Hepatitis B birth-dose vaccine (± HBIG)	Standard dose	IM	Once	Within 24 hours of birth	Then pentavalent series

Under 2.0 kg: mg/kg dosing per the national paediatric chart; discuss with the paediatrician.

Referral criteria

Level	What should happen	Refer up when
HC II / outreach	Counselling, HIV/syphilis testing, referral/linkage; PrEP refill in some programmes	Every new positive → HC III for ART initiation
HC III	Full eMTCT: testing, TLD initiation, CTX, NVP, EID sample collection (DBS), VL sample collection via hub	Danger signs (breathless, headache with neck stiffness, jaundice, high fever), CD4 below 200 needing CrAg/LP, obstetric complications needing CS → HC IV
HC IV	As HC III, plus CS for obstetric indications, POC VL/EID in many sites, management of complications, AZT + NVP for high-risk infants	Advanced HIV disease needing LP, suspected cryptococcal or TB meningitis, persistent non-suppression for the switch committee, renal impairment, HBV with abnormal LFTs → General Hospital / RRH
General / Regional Referral Hospital	AHD management (CrAg, LP, TB LAM), switch committee for treatment failure, resistance testing, neonatal unit, complex co-morbidities (TB, HBV, renal disease)	Drug-resistant TB, complex resistance, HBV DNA testing, specialist care → NRH
National Referral Hospital	Specialist HIV, hepatitis, TB and neonatal care; HBV DNA	n/a

Monitoring and follow-up

- **Every ANC visit**: adherence, TB screen, side effects, weight, BP, SFH, fetal heart, CTX refill (table above).
- **Viral load**: first ANC visit (if on ART) or 3 months after start, then every 3 months until breastfeeding ends; recent result at 34–36 weeks.
- **Infant**: prophylaxis adherence; EID at 4–6 weeks, 9 months, 6 weeks after weaning, antibody at 18 months; co-trimoxazole from 6 weeks; growth and immunisation.
- **Postpartum**: VL every 3 months while breastfeeding, then every 6–12 months; family planning; cervical screening; mother-baby pair until final status.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Opportunistic infections: TB, PCP, cryptococcal meningitis	**HIV infection** (in utero, intrapartum, breastfeeding)
Anaemia	Preterm birth
Malaria (more frequent and severe, placental malaria)	Low birth weight / growth restriction
Puerperal sepsis, post-caesarean wound infection	Stillbirth

Maternal	Fetal / neonatal
Depression, stigma, intimate partner violence after disclosure	Congenital syphilis, hepatitis B (co-infections)
Loss to follow-up postpartum	Missed EID and late diagnosis
	HIV-exposed-uninfected infants: slightly higher morbidity

Treatment-related

Complication	Cause	Recognition and action
Renal impairment	TDF	Creatinine; switch to ABC-based regimen (check HBsAg first)
Neuropsychiatric effects (vivid dreams, dizziness)	EFV	Take at night; switch to DTG
Hepatotoxicity	ART, TB drugs, co-trimoxazole	LFTs; senior advice before stopping drugs
Weight gain	DTG	Counsel; monitor
Immune reconstitution inflammatory syndrome (IRIS)	ART start in AHD	Delay ART in cryptococcal (4–6 weeks) and TB meningitis
Severe skin reactions, kernicterus risk	Co-trimoxazole combined with IPTp-SP	Never combine
HBV flare	Stopping TDF in HBV co-infection	Do not stop TDF abruptly
Infant anaemia, neutropenia	AZT	FBC if pale

11. Prognosis

- **For the baby**: with a suppressed maternal VL throughout pregnancy and breastfeeding, transmission is **below 1%**; without intervention it is 15–45%.
- **For the mother**: on suppressive lifelong ART, a woman with HIV can expect a long life Verify *no outcome figure given in the source notes*. Outcomes are worse with advanced HIV disease, TB and cryptococcal meningitis, and with disengagement from care postpartum.
- **HIV-exposed-uninfected infants** have slightly higher morbidity than unexposed infants.

12. Prevention

eMTCT is organised in four prongs (WHO framework) Verify *Uganda programme wording*:

1. **Primary prevention of HIV in women of reproductive age**: testing, condoms, **PrEP** for women at substantial risk, partner testing, **retesting** in the third trimester, in labour if missed, and during breastfeeding.
2. **Prevention of unintended pregnancy among women with HIV**: family planning (DTG has no interaction with hormonal contraceptives; EFV lowers implant levels).
3. **Prevention of transmission from mother to child**: same-day TLD, VL monitoring, safe intrapartum practice, infant prophylaxis, exclusive breastfeeding.
4. **Care and support for the mother, child and family**: lifelong ART, EID, co-trimoxazole, differentiated service delivery, the mother-baby pair.

Triple elimination: syphilis (dual RDT and benzathine penicillin) and hepatitis B (HBsAg, TDF from about 28 weeks, birth-dose vaccine within 24 hours).

13. Special Populations

- **Newly diagnosed in labour or postpartum**: TLD now; baby high-risk (AZT + NVP within 6 hours); HBsAg and syphilis tests; link mother and baby to the eMTCT clinic.
- **Diagnosed while breastfeeding**: mother on TLD, baby AZT + NVP, **DNA PCR now**, high-risk for 12 weeks.
- **HIV/HBV co-infection**: TLD treats both; never an ABC-based regimen alone; do not stop TDF abruptly; HBV birth dose within 24 hours.
- **HIV/TB co-infection**: TB treatment first, ART within 2 weeks; **DTG 50 mg twice daily** with rifampicin and for 2 weeks after; TPT only after TB excluded.
- **Advanced HIV disease / cryptococcal meningitis**: CrAg, TB LAM; delay ART 4–6 weeks after antifungals; fluconazole in the first trimester needs senior advice.
- **Renal impairment** (CrCl below 50 mL/min): ABC + 3TC + DTG; check HBsAg first.
- **Serodiscordant couple**: partner on PrEP if the woman is not suppressed; HIV-negative woman with a positive partner gets PrEP and retesting.
- **Adolescent mothers**: young women aged 15–24 carry a disproportionate share of new infections; disclosure, adherence and retention are hardest Verify *add adolescent-specific Uganda service guidance*.

14. Key Points

High-yield points

- Viral load is the single biggest determinant of transmission: MTCT 15–45% untreated, below 1% if suppressed.
- Test (HIV, syphilis, HBsAg) at booking and **retest** negatives.
- Same-day **TLD** for every pregnant or breastfeeding woman with HIV.
- VL at first ANC (if on ART) or 3 months after start, then every 3 months until weaning.
- Present the **four facts** and the **infant risk category**.
- High-risk infant: AZT + NVP 6 weeks, then NVP 6 weeks.
- PCR, not antibody, before 18 months.
- Exclusive breastfeeding for 6 months on ART; never mix-feed.
- Co-trimoxazole, so no IPTp-SP.
- CS only for obstetric indications.

Numbers & doses

Item	Value
MTCT risk, no intervention / suppressed on ART	**15–45%** / **below 1%**
TLD	TDF 300 mg + 3TC 300 mg + DTG 50 mg, **one tablet once daily**, start **same day**
DTG with rifampicin	**50 mg twice daily**, until **2 weeks after** rifampicin stops
DTG with iron/calcium/antacids	DTG **2 h before or 6 h after**, or together with food
VL schedule	First ANC visit (if on ART) or **3 months after start**, then **every 3 months** until breastfeeding ends
VL categories	Below 50 undetectable; 50–999 low-level viraemia; **1,000 or more non-suppressed** (programme cut-off; check local rule)
Advanced HIV disease	**CD4 below 200** or WHO stage 3–4

Item	Value
Delay ART in cryptococcal meningitis	**4–6 weeks** after starting antifungals
Co-trimoxazole (mother)	**960 mg once daily**; no IPTp-SP
Isoniazid preventive therapy	**300 mg daily × 6 months + pyridoxine 25 mg daily** (TB excluded first)
Infant NVP / AZT, 2.5 kg or more	**15 mg (1.5 mL)** NVP once daily / **15 mg (1.5 mL)** AZT twice daily
Infant NVP / AZT, 2.0–2.49 kg	**10 mg (1 mL)** each
Infant NVP 6–12 weeks	**20 mg (2 mL)** once daily
Start infant prophylaxis	**Within 6 hours** of birth
Low-risk / high-risk infant prophylaxis	NVP **6 weeks** / AZT + NVP 6 weeks then NVP 6 weeks (**12 weeks**)
High-risk definition	ART **under 4 weeks**, VL **1,000 or more or unknown** in the last 4 weeks, late diagnosis, seroconversion
EID	DNA PCR **4–6 weeks, 9 months, 6 weeks after weaning**; antibody at **18 months**
Infant co-trimoxazole	From **6 weeks**: **120 mg** daily under 6 months, **240 mg** from 6 months
Syphilis	Benzathine penicillin **2.4 MIU IM** single (early) or **weekly × 3** (late/unknown)
HBsAg-positive, HIV-negative	TDF **300 mg daily from about 28 weeks**; baby HBV vaccine **within 24 hours**
PEP after needlestick	TLD for **28 days**

Exam traps

- **"Give IPTp-SP as usual."** Wrong: a woman on co-trimoxazole gets **no SP**. Co-trimoxazole protects against malaria; give an ITN.
- **"Do an HIV antibody test at 6 weeks."** Wrong: maternal IgG persists to about 18 months. Use **DNA PCR**.
- **"Plan an elective caesarean because she is HIV-positive."** Wrong in Uganda: CS is for **obstetric indications**.
- **"Switch to second line because the VL is 2,400."** Wrong first step: do **IAC and repeat the VL**; most non-suppression is adherence.
- **"She was suppressed last year, so the baby is low-risk."** Wrong: risk depends on a VL **in the last 4 weeks before delivery**; old or unknown VL = **high-risk**.
- **"Formula-feed to protect the baby."** Wrong in Uganda: **exclusive breastfeeding for 6 months** on ART; never mix-feed.
- **"Start ART today" in a woman with headache and neck stiffness.** Wrong: rule out cryptococcal/TB meningitis first; ART is **delayed 4–6 weeks** after antifungals start.
- **"Stop DTG in women who might conceive."** Wrong: WHO and Uganda recommend DTG for all, including pregnant women, with folic acid as usual.

Questions seniors ask

On the ward round

1. **"What was her last viral load, and when?"** The one answer that decides the baby's prophylaxis. If it is 1,000 or more, unknown, or not from the last 4 weeks, the baby is high-risk.
2. **"Why is TLD preferred in pregnancy?"** DTG suppresses the VL quickly (vital in late presenters), has a high genetic barrier to resistance, is well tolerated, and TDF + 3TC also treat hepatitis B.

3. **"She is on TLD and ferrous sulphate and her VL is up. What will you check?"** Adherence first, then **timing of iron with DTG** (chelation), then other interactions such as rifampicin; resistance is uncommon on DTG.
4. **"The baby weighs 2.7 kg and the mother started ART 2 weeks ago. Prescribe."** High-risk: **AZT 15 mg (1.5 mL) twice daily + NVP 15 mg (1.5 mL) once daily for 6 weeks**, then **NVP 20 mg once daily** for another 6 weeks.
5. **"When is the baby's HIV status final?"** After a negative test done **at least 6 weeks after all breastfeeding has stopped** (and the 18-month antibody test).
6. **"Her dual RDT shows syphilis positive, but she says she was treated last pregnancy. What now?"** The treponemal RDT stays positive for life; if treatment is documented and adequate and there is no new exposure, do not simply re-treat. If undocumented or doubtful, treat as unknown duration: **benzathine penicillin 2.4 MIU weekly × 3**.
7. **"Her CD4 is 150. What extra tests?" Serum CrAg** and **urine TB LAM**, a TB symptom screen, GeneXpert if symptomatic; she has advanced HIV disease.
8. **"She presents in labour with unknown status. What do you do?"** Rapid HIV test now; if positive, **TLD immediately**, baby **high-risk** (AZT + NVP within 6 hours), HBsAg and syphilis tests, and link mother and baby to the eMTCT clinic.

Viva questions

1. **What are the periods of MTCT and their relative contribution?** In utero 5–10%, intrapartum 10–20%, breastfeeding 5–20% (untreated). Intrapartum is the biggest single window.
2. **What is Option B+?** Lifelong ART for all pregnant and breastfeeding women with HIV regardless of CD4, adopted by Uganda in 2012; now part of universal test-and-treat.
3. **What is the first-line regimen and its dose?** TLD: TDF 300 mg, 3TC 300 mg, DTG 50 mg, one tablet once daily.
4. **How do you define a high-risk infant?** Mother diagnosed late in pregnancy, in labour or while breastfeeding; less than 4 weeks of ART before delivery; VL of 1,000 or more (or unknown) in the 4 weeks before delivery; seroconversion during pregnancy or breastfeeding.
5. **What prophylaxis does a high-risk infant get?** AZT + NVP for 6 weeks, then NVP for 6 more weeks.
6. **Why can't you use an antibody test at 6 weeks?** Maternal IgG crosses the placenta and persists up to 18 months; DNA PCR detects the virus itself.
7. **Why no IPTp-SP with co-trimoxazole?** Overlapping sulfonamide toxicity (SJS, kernicterus) without added benefit; CTX already protects against malaria.
8. **What do you do with a VL of 4,500 copies/mL at 28 weeks?** IAC for 3 sessions, look for interactions, repeat VL; baby high-risk if not suppressed near delivery; switch committee if persistent despite good adherence.
9. **What intrapartum practices reduce transmission?** Continue ART, avoid prolonged ROM, unnecessary ARM, episiotomy, scalp electrodes, prolonged labour; prophylaxis to baby within 6 hours.
10. **DTG with TB treatment?** DTG 50 mg twice daily during rifampicin and for 2 weeks after.

Mnemonics

"TEST, TREAT, TRACK, PROTECT"

- **TEST**: HIV + syphilis + HBsAg at booking; retest negatives.
- **TREAT**: TLD same day; CTX (no SP); TPT once TB excluded.
- **TRACK**: VL at first ANC (or 3 months after start), then every 3 months until weaning.
- **PROTECT**: infant NVP ± AZT, PCR at 6 weeks / 9 months / 6 weeks after weaning, antibody at 18 months, exclusive breastfeeding.

High-risk infant: "LATE, LOW, HIGH, NEW"

- **LATE**: diagnosed late pregnancy, labour or breastfeeding
- **LOW**: less than 4 weeks of ART
- **HIGH**: VL 1,000 or more (or unknown) near delivery
- **NEW**: seroconverted in pregnancy/breastfeeding

EID: "6 weeks, 9 months, 6 weeks after, 18 months", a **6-9-6-18** rhythm.

DTG interactions: "RIM": **R**ifampicin (double DTG), **I**ron/calcium/antacids (separate by 2 h before / 6 h after), **M**etformin (lower dose).

The infant-risk decision tree is drawn in section 9 (Infant prophylaxis).

OSCE tips

- In a **counselling station** (new positive result in ANC): check readiness, give the result clearly, allow silence and emotion, reassure that with treatment her baby can be born HIV-free, explain same-day ART and adherence, discuss disclosure and partner testing, safety, next steps. Avoid jargon; check understanding with teach-back.
- In an **infant prophylaxis station**: state weight, risk category, drug, dose in mg **and** mL, frequency, duration, and demonstrate syringe measurement.
- In a **history station**: get the four facts (regimen, time on ART, last VL with date, adherence), the TB screen, co-trimoxazole, triple elimination results, and the birth and feeding plan.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What is the approximate risk of mother-to-child transmission of HIV without any intervention, and with a suppressed maternal viral load?	**15–45%** without intervention (higher with prolonged breastfeeding). **Below 1%** when the mother is on ART with a suppressed viral load throughout pregnancy and breastfeeding.
2	Name the three periods of MTCT and their approximate contribution (untreated).	**In utero** 5–10%, **intrapartum** 10–20% (largest single window), **breastfeeding** 5–20%.
3	What is 'triple elimination'?	Integrated elimination of vertical transmission of **HIV, syphilis and hepatitis B**: dual HIV/syphilis test + **HBsAg** at the first ANC visit, with treatment and infant prevention for each.
4	What is Uganda's preferred first-line ART for a pregnant woman, and its dose?	**TLD**: tenofovir DF 300 mg + lamivudine 300 mg + dolutegravir 50 mg, **one tablet orally once daily**, started the same day as diagnosis where possible.
5	When should ART be started in a newly diagnosed pregnant woman?	**The same day** (rapid initiation), after counselling, unless cryptococcal or TB meningitis is suspected (then defer ART by about 4–6 weeks after starting treatment).
6	What is the viral load monitoring schedule for pregnant and breastfeeding women in Uganda?	Already on ART: VL at the **first ANC visit**. Newly started: VL **3 months after starting**. Then **every 3 months until breastfeeding ends**.
7	What VL defines non-suppression and what is the first response?	**1,000 copies/mL or more**. Respond with **intensive adherence counselling** (about 3 monthly sessions), check interactions, then **repeat VL**. Switch only via the switch committee if still non-suppressed with good adherence.
8	Define a high-risk HIV-exposed infant.	Mother: diagnosed in **late pregnancy, labour or breastfeeding**; on ART **less than 4 weeks** before delivery; **VL 1,000 or more (or unknown)** in the 4 weeks before delivery; or **seroconverted** during pregnancy/breastfeeding.
9	Prophylaxis for a LOW-risk HIV-exposed infant?	**Nevirapine once daily for 6 weeks**: 10 mg (1 mL) if 2.0–2.49 kg, 15 mg (1.5 mL) if 2.5 kg or more. Start within 6 hours of birth.
10	Prophylaxis for a HIGH-risk HIV-exposed infant?	**AZT twice daily + NVP once daily for 6 weeks**, then **NVP alone for another 6 weeks** (12 weeks total). Extend NVP if the mother's VL is still 1,000 or more at about 12 weeks postpartum.
11	Infant NVP and AZT doses for a 3.1 kg newborn?	NVP **15 mg (1.5 mL of 10 mg/mL) once daily**; AZT **15 mg (1.5 mL of 10 mg/mL) twice daily**. From 6 to 12 weeks, NVP 20 mg once daily.
12	Uganda EID schedule?	DNA PCR at **4–6 weeks**, **9 months**, and **6 weeks after complete cessation of breastfeeding**; **rapid antibody test at 18 months**. Any positive PCR: start ART and send a confirmatory DBS.
13	Why is an HIV antibody test not used to diagnose infection in a 6-week-old?	**Maternal IgG antibodies cross the placenta** and can persist up to about 18 months, so the baby tests antibody-positive whether infected or not. DNA PCR detects the virus itself.
14	Why must a woman on co-trimoxazole not receive IPTp-SP?	Both are **sulfonamide antifolates**: combining adds toxicity (Stevens-Johnson syndrome, kernicterus) with no extra benefit. Co-trimoxazole already protects against malaria.

#	Question	Answer
15	How does rifampicin affect dolutegravir, and what do you do?	Rifampicin **induces** DTG metabolism and lowers levels. Give **DTG 50 mg twice daily** (TLD + extra DTG 50 mg) throughout rifampicin and for **2 weeks after** stopping it.
16	How should DTG be timed with ferrous sulphate or calcium?	Take DTG **2 hours before or 6 hours after** iron, calcium, zinc or antacids, or take them together **with food**. Polyvalent cations chelate DTG and reduce absorption.
17	Uganda infant feeding recommendation for an HIV-positive mother on ART?	**Exclusive breastfeeding for 6 months**, then complementary foods with **continued breastfeeding to at least 12 months** (up to 24), stopping gradually. **No mixed feeding** in the first 6 months.
18	Why does mixed feeding increase HIV transmission?	Non-breast-milk foods and fluids **inflame and damage the infant's gut mucosa**, letting HIV in breast milk cross the gut wall. Exclusive breastfeeding keeps the barrier intact.
19	Treatment of syphilis detected at ANC (unknown duration)?	**Benzathine penicillin G 2.4 MIU IM weekly for 3 doses** (single dose if early syphilis). Treat the partner. If penicillin-allergic, erythromycin 500 mg 6-hourly for 14 days and treat the baby after birth.
20	PrEP in pregnancy: who, what, how monitored?	HIV-negative pregnant/breastfeeding women at substantial risk (e.g. partner not suppressed). **Oral TDF/FTC (or TDF/3TC) once daily**, safe in pregnancy and breastfeeding. **HIV test every 3 months**; creatinine where available.
21	Intrapartum practices to avoid in an HIV-positive woman?	Prolonged rupture of membranes, **routine ARM**, **routine episiotomy**, **fetal scalp electrodes/blood sampling**, unnecessary instrumental delivery, prolonged labour, excessive VEs. Continue her ART in labour.
22	Is caesarean section routinely indicated for HIV in Uganda?	**No.** CS is for obstetric indications. Vaginal delivery with ART and infant prophylaxis is standard; CS carries risks of sepsis and haemorrhage in our setting.
23	What four facts must you state when presenting any HIV-positive pregnant woman on a ward round?	**Regimen**, **time on ART**, **last viral load with its date**, and **adherence**. Then state the **infant risk category** (low or high).
24	Which bedside tests identify advanced HIV disease and its main killers on the same day?	**Point-of-care CD4** (below 200 = AHD), **serum CrAg lateral-flow test** (cryptococcal antigen, if CD4 200 or below) and **urine TB LAM** (CD4 below 200 or seriously ill).
25	A woman on TLD has a VL of 2,400 copies/mL at 33 weeks. Give the next steps.	**IAC** (starting today), check **iron/calcium timing with DTG** and other interactions (rifampicin), **repeat VL early** so a result is available before delivery, and plan **high-risk infant prophylaxis** (AZT + NVP 6 weeks, then NVP 6 weeks) unless a VL below 1,000 is available from the last 4 weeks before delivery. Do **not** switch regimen as the first step.
26	Why does a positive dual HIV/syphilis RDT not always mean active syphilis?	The syphilis component is **treponemal**, so it **stays positive for life** after treatment. Ask about documented past treatment. If undocumented or doubtful, treat as unknown duration: **benzathine penicillin 2.4 MIU IM weekly × 3**.
27	A woman newly diagnosed with HIV in labour is also HBsAg-positive. What does the baby need?	**High-risk HIV prophylaxis**: AZT + NVP within 6 hours of birth (12 weeks in total), and the **hepatitis B birth-dose vaccine within 24 hours** (± HBIG where available). The mother starts **TLD now** (TDF + 3TC also treat HBV).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 24-year-old G2P1 at 16 weeks books at a HC IV. Her HIV rapid test is positive and is confirmed on the national algorithm. She is well, with no headache, cough or fever. What is the MOST appropriate next step?

A Send CD4 count and start ART when the result returns in 2 weeks
B Start TLD (TDF/3TC/DTG) one tablet once daily today after counselling
C Start zidovudine monotherapy from 28 weeks
D Defer ART until after the first trimester has been confirmed by ultrasound
E Refer to the regional referral hospital for ART initiation

2. A 30-year-old woman on TLD for 4 years is at 20 weeks. She attends ANC and receives ferrous sulphate/folic acid, and is started on co-trimoxazole. The midwife prepares IPTp-SP. What is the BEST action?

A Give IPTp-SP as scheduled monthly from 13 weeks
B Give IPTp-SP but stop co-trimoxazole during pregnancy
C Give IPTp-SP and co-trimoxazole on alternate days
D Replace co-trimoxazole with weekly chloroquine
E Withhold IPTp-SP; continue co-trimoxazole and give an ITN

3. A 27-year-old P1 delivers a 2.8 kg baby at 39 weeks. She was diagnosed with HIV and started TLD 2 weeks before delivery. What prophylaxis should the baby receive?

A NVP 15 mg once daily for 6 weeks
B NVP 10 mg once daily for 6 weeks
C AZT 15 mg twice daily + NVP 15 mg once daily for 6 weeks, then NVP alone for a further 6 weeks
D No prophylaxis because the mother is now on ART
E Single-dose NVP at birth only

4. A 32-year-old at 30 weeks has been on TLD for 3 years. Her VL at booking (18 weeks) was 3,800 copies/mL. She admits missing doses when her husband is home because she has not disclosed. What is the MOST appropriate management now?

A Start intensive adherence counselling, address disclosure, and repeat the VL after completing the sessions
B Switch immediately to a second-line protease-inhibitor regimen
C Add efavirenz to her current regimen
D Stop ART until she has disclosed, to prevent resistance
E Plan an elective caesarean section at 36 weeks and continue the current approach

5. A 25-year-old at 34 weeks on TLD is diagnosed with smear-negative pulmonary TB (GeneXpert positive, rifampicin sensitive) and started on RHZE. What change to her ART is required?

A No change
B Switch TLD to TLE400
C Stop ART until TB treatment is complete
D Add an extra dolutegravir 50 mg so that DTG is taken twice daily, continued for 2 weeks after rifampicin stops
E Double the whole TLD tablet to two tablets twice daily

6. A 23-year-old unbooked woman arrives at a HC IV in active labour, 7 cm dilated, membranes intact. A rapid HIV test is positive. What is the MOST appropriate plan?

A Perform an emergency caesarean section to reduce transmission

B Start TLD now, avoid artificial rupture of membranes and episiotomy, and give the baby AZT + NVP within 6 hours of birth

C Give the mother single-dose nevirapine only

D Wait for a CD4 count before any ART

E Give the baby NVP alone for 6 weeks and start the mother on ART at the postnatal visit

7. A mother brings her 6-week-old HIV-exposed baby for the first immunisations. The baby is exclusively breastfed and completed NVP prophylaxis. Which test should be done today?

A Rapid HIV antibody test

B No test until 18 months

C CD4 percentage

D HIV viral load on the baby

E HIV DNA PCR on a dried blood spot or point-of-care platform

8. A 28-year-old HIV-positive mother on TLD with a suppressed VL asks how she should feed her newborn. She lives in a village with an unprotected water source. What is the BEST advice?

A Formula feed exclusively from birth

B Breastfeed and give water and glucose between feeds during hot weather

C Exclusive breastfeeding for 6 months, then complementary foods with continued breastfeeding to at least 12 months while continuing ART

D Breastfeed for 6 weeks then stop abruptly

E Express and heat-treat breast milk for 2 years

9. A HIV-exposed infant was on AZT + NVP for 6 weeks then NVP for 6 more because the mother's VL at delivery was 9,200 copies/mL. At 12 weeks postpartum the mother's VL is 2,400 copies/mL. She is breastfeeding. What should happen to the infant's prophylaxis? Advanced

A Continue NVP until the mother's VL is suppressed (or until 4 weeks after breastfeeding stops)

B Stop NVP now as 12 weeks are complete

C Restart AZT + NVP for another 12 weeks

D Switch the infant to co-trimoxazole alone

E Stop breastfeeding and stop NVP

10. A 21-year-old HIV-negative woman at 26 weeks reports her husband is HIV-positive and has 'not been taking his drugs for months'. What is the MOST appropriate intervention for her?

A Reassure her; pregnancy reduces the risk of HIV acquisition

B Advise her to avoid breastfeeding

C Start her on TLD immediately

D Offer oral PrEP (TDF/FTC once daily) after a negative HIV test, with retesting every 3 months

E Test her again only at delivery

11. A 29-year-old at 30 weeks is HIV-negative, syphilis-negative, but HBsAg-positive. HBV DNA is 3.5 million IU/mL. What is the MOST appropriate management to reduce vertical transmission? Advanced

A Hepatitis B vaccine to the mother only

B Tenofovir DF 300 mg once daily now, continuing until at least delivery (often to 12 weeks postpartum), plus the infant HBV birth-dose vaccine within 24 hours (± HBIG)

C Elective caesarean section at 38 weeks as the main intervention

D Avoid breastfeeding

E Lamivudine monotherapy from 36 weeks

12. A 26-year-old on TLD at 37 weeks has a VL of below 50 copies/mL done 1 month ago. She has spontaneous rupture of membranes with no contractions. Cervix is unfavourable. There are no other complications. What is the BEST plan? Advanced

A Expectant management for up to 72 hours

B Emergency caesarean section because she is HIV-positive

C Discharge home to await labour

D Give IV zidovudine infusion and wait

E Expedite delivery: induce labour promptly (e.g. misoprostol per protocol), continue ART and avoid unnecessary vaginal examinations

13. A 34-year-old HIV-positive woman on TLD attends the 6-week postnatal clinic. She wants a contraceptive implant. Which statement is correct?

A Implants are contraindicated with any ART

B She must switch to efavirenz before the implant is inserted

C DTG does not reduce implant efficacy; the implant is a suitable choice

D She should use condoms only until breastfeeding stops

E DTG and levonorgestrel implant together cause neural tube defects

14. A 22-year-old tested HIV-negative at booking (12 weeks). At 32 weeks she presents with fever, sore throat, a maculopapular rash and generalised lymphadenopathy. Malaria RDT is negative. What is the MOST important test to request? Advanced

A Repeat HIV test (and HIV RNA/VL if the rapid test is negative and suspicion persists)

B Monospot test only

C Blood culture only

D Throat swab for streptococcus

E Antinuclear antibodies

15. A 3.0 kg baby is born to a mother on TLD for 6 years with a VL below 50 copies/mL 2 months ago. What is the correct prophylaxis?

A NVP 10 mg once daily for 6 weeks

B NVP 20 mg once daily for 12 weeks

C AZT + NVP for 12 weeks

D NVP 15 mg (1.5 mL) once daily for 6 weeks

E No prophylaxis as the VL is suppressed

Answers & explanations

1. B · Start TLD (TDF/3TC/DTG) one tablet once daily today after counselling

B is correct. Uganda practises test-and-treat with **rapid (same-day) ART initiation**; TLD is the preferred first-line regimen in pregnancy. **A**: baseline CD4 is useful (to identify advanced disease) but ART must not wait for it. **C**: AZT monotherapy from 28 weeks is an obsolete 'Option A'-era approach. **D**: she is already in the second trimester and DTG is recommended in all trimesters; delay increases transmission risk. **E**: HC IVs (and HC IIIs) initiate ART; referral wastes time and risks loss to follow-up.

2. E · Withhold IPTp-SP; continue co-trimoxazole and give an ITN

E is correct. Women on co-trimoxazole **must not receive IPTp-SP**: both are sulfonamide antifolates, so combining them adds toxicity (SJS, kernicterus) without benefit; CTX itself protects against malaria. **A** and **C** combine the two. **B**: CTX also prevents PCP, toxoplasmosis and bacterial infections in PLHIV; it should not be stopped for SP. **D**: chloroquine prophylaxis is not used because of resistance.

3. C · AZT 15 mg twice daily + NVP 15 mg once daily for 6 weeks, then NVP alone for a further 6 weeks

C is correct. The mother received **less than 4 weeks of ART** before delivery, so the infant is **high-risk**: AZT + NVP for 6 weeks, then NVP for 6 more weeks (12 total). **A** is the low-risk regimen. **B** is also the low-risk regimen and the wrong weight band (10 mg is for 2.0–2.49 kg). **D**: maternal ART for 2 weeks has not suppressed the VL. **E**: single-dose NVP is obsolete and inadequate for breastfeeding exposure.

4. A · Start intensive adherence counselling, address disclosure, and repeat the VL after completing the sessions

A is correct. Non-suppression is most often due to **adherence**; here the barrier (non-disclosure) is obvious. IAC (around 3 sessions), disclosure support and a treatment supporter, then **repeat VL**. **B**: switching without addressing adherence leads to failure of the new regimen, and switches go through the switch committee after a repeat VL. **C**: adding a single drug to a failing regimen breeds resistance. **D**: stopping ART raises VL and transmission risk. **E**: CS is not the answer; suppression is. The infant will be classified high-risk if she is still not suppressed near delivery.

5. D · Add an extra dolutegravir 50 mg so that DTG is taken twice daily, continued for 2 weeks after rifampicin stops

D is correct. Rifampicin induces DTG metabolism, so DTG must be given **50 mg twice daily** during rifampicin and for **2 weeks after**. **A** risks subtherapeutic DTG and resistance. **B**: switching to EFV is unnecessary and EFV is a weaker regimen. **C**: stopping ART in an HIV/TB co-infected pregnant woman increases mortality and MTCT. **E**: doubling TDF and 3TC adds toxicity with no benefit; only DTG needs doubling.

6. B · Start TLD now, avoid artificial rupture of membranes and episiotomy, and give the baby AZT + NVP within 6 hours of birth

B is correct. Diagnosis in labour means a **high-risk infant**: start the mother on TLD immediately, minimise intrapartum exposure (avoid ARM, episiotomy, scalp electrodes, prolonged labour), and give the baby **AZT + NVP** (then NVP to 12 weeks). **A**: at 7 cm, CS offers little benefit and adds maternal risk; not policy in Uganda. **C**: single-dose NVP is obsolete. **D**: never delay ART for CD4. **E**: under-treats a high-risk baby and delays maternal ART.

7. E · HIV DNA PCR on a dried blood spot or point-of-care platform

E is correct. The first **DNA PCR** is due at 4–6 weeks. **A**: maternal antibodies persist until up to 18 months, so an antibody test cannot diagnose infection now. **C**: CD4 is not diagnostic. **D**: VL is used for monitoring treatment, not first-line EID. **B**: waiting would miss infected infants, who have high early mortality without ART.

8. C · Exclusive breastfeeding for 6 months, then complementary foods with continued breastfeeding to at least 12 months while continuing ART

C is correct (Uganda/WHO). **A**: formula is not AFASS here (unsafe water, cost), so diarrhoea and malnutrition risks outweigh the small transmission risk from a suppressed mother. **B**: mixed feeding damages the infant gut mucosa and increases transmission. **D**: abrupt early cessation deprives the baby of nutrition and immunity and is not recommended. **E**: heat-treating is impractical and not the standard recommendation.

9. A · Continue NVP until the mother's VL is suppressed (or until 4 weeks after breastfeeding stops)

A is correct. Uganda guidance extends **NVP** beyond 12 weeks while the breastfeeding mother remains non-suppressed, until she suppresses or until 4 weeks after breastfeeding ends. **B** leaves the infant unprotected during ongoing high-level exposure. **C**: dual therapy is not required beyond the first 6 weeks. **D**: CTX prevents opportunistic infections but not HIV. **E**: abrupt cessation is harmful and does not replace fixing maternal suppression (IAC, repeat VL, switch committee).

10. D · Offer oral PrEP (TDF/FTC once daily) after a negative HIV test, with retesting every 3 months

D is correct. She is at substantial risk; **oral PrEP (TDF/FTC or TDF/3TC)** is safe in pregnancy and breastfeeding. HIV test before starting and every 3 months; also support the partner's re-engagement. **A** is false: pregnancy and postpartum **increase** acquisition risk. **C**: full ART is for people with HIV; she is negative. **B**: she is negative; breastfeeding is recommended. **E**: she needs prevention now and retesting per the schedule, not just at delivery.

11. B · Tenofovir DF 300 mg once daily now, continuing until at least delivery (often to 12 weeks postpartum), plus the infant HBV birth-dose vaccine within 24 hours (± HBIG)

B is correct. High HBV DNA (200,000 IU/mL or more) warrants **TDF prophylaxis from about 28 weeks**, plus a **timely birth-dose vaccine** (and HBIG where available). **A**: vaccinating an infected mother is useless. **C**: CS is not recommended for HBV prevention. **D**: breastfeeding is safe with infant vaccination. **E**: TDF is preferred; 3TC alone has a low resistance barrier and starting at 36 weeks is late.

12. E · Expedite delivery: induce labour promptly (e.g. misoprostol per protocol), continue ART and avoid unnecessary vaginal examinations

E is correct. In HIV-positive women with term PROM, **delivery should be expedited** to shorten exposure; with a suppressed VL, vaginal birth is appropriate. **A** and **C** prolong exposure and infection risk. **B**: CS is not indicated for HIV alone with a suppressed VL. **D**: IV AZT is not part of Uganda's intrapartum protocol and does not replace expediting delivery.

13. C · DTG does not reduce implant efficacy; the implant is a suitable choice

C is correct. DTG has **no clinically significant interaction** with hormonal contraceptives. **A**: false. **B**: it is **efavirenz** that reduces implant levels (the opposite of helpful). **D**: dual protection is sensible, but she need not rely on condoms alone. **E**: contraceptives prevent pregnancy; the DTG NTD signal has largely resolved and does not relate to implants.

14. A · Repeat HIV test (and HIV RNA/VL if the rapid test is negative and suspicion persists)

A is correct. This is a classic **acute retroviral syndrome**; seroconversion in pregnancy carries a very high MTCT risk because of the huge VL. Antibody tests may still be negative in the window period, so check **HIV RNA** if available. **B**, **C**, **D** consider other causes but miss the most dangerous one. **E** is irrelevant here. This case illustrates why **retesting** is part of eMTCT.

15. D · NVP 15 mg (1.5 mL) once daily for 6 weeks

D is correct. Long-term ART with a recent suppressed VL makes the infant **low-risk**: **NVP once daily for 6 weeks**, 15 mg (1.5 mL) for 2.5 kg or more. **B**: 20 mg is the 6–12-week dose and 12 weeks is for high-risk infants. **C**: the high-risk regimen. **A**: 10 mg is for 2.0–2.49 kg. **E**: all HEIs in Uganda receive prophylaxis because breastfeeding exposure continues.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: New HIV diagnosis at booking

Aisha, 22, G1P0 at 14 weeks, books at a HC IV in Wakiso. Her dual HIV/syphilis test is HIV-positive, syphilis-negative. She is tearful and says her boyfriend does not know her status. She looks well.

1. What history do you take now, and why?

> Confirm the result (national serial algorithm). Ask about **symptoms of TB** (cough, fever, night sweats, poor weight gain), **headache/neck stiffness** (cryptococcal meningitis would delay ART), rash/jaundice, other medicines (TB drugs, anticonvulsants), previous HIV tests, partner status and **safety at home** (intimate partner violence risk on disclosure), mental health, support network, and her plans for delivery and feeding.

2. What examination and investigations?

> Weight, BP, pallor, lymph nodes, oral thrush, skin (KS, zoster), chest, neck stiffness; obstetric exam. Tests: **CD4**, HBsAg, Hb, creatinine if available, urinalysis, TB symptom screen (GeneXpert if symptomatic), **CrAg if CD4 200 or below**. ART is **not delayed** for results.

3. What treatment do you start today?

> After counselling: **TLD one tablet once daily** (TDF 300 / 3TC 300 / DTG 50), same day. **Co-trimoxazole 960 mg once daily** (and **no IPTp-SP**). Continue ANC ferrous sulphate/folic acid, **timed 2 h after or 6 h before DTG** or taken with food. ITN. TPT (isoniazid 300 mg + pyridoxine 25 mg daily for 6 months) once TB is excluded. Identify a treatment supporter; plan disclosure and partner testing (assisted partner notification).

4. How will you monitor her during pregnancy?

> Monthly (or per schedule) adherence checks, TB screen every visit, side effects (DTG: insomnia, headache). **VL 3 months after starting** (about 26 weeks), then every 3 months until breastfeeding ends. Ensure a VL result is available around 34–36 weeks to classify the baby's risk.

5. Her VL at 26 weeks is below 50 copies/mL. She delivers a 2.4 kg baby at 38 weeks. What does the baby receive?

> She has been on ART for more than 4 weeks and is suppressed, so the baby is **low-risk**: **NVP 10 mg (1 mL) once daily for 6 weeks** (2.0–2.49 kg band), started within 6 hours. Breastfeed within 1 hour, exclusively for 6 months. Vitamin K, BCG, OPV-0. **Co-trimoxazole from 6 weeks. DNA PCR at 6 weeks, 9 months, 6 weeks after weaning; antibody at 18 months.**

6. What postnatal counselling does she need?

> Lifelong ART, VL every 3 months while breastfeeding, family planning (DTG has no interaction with implants/DMPA), cervical screening, mother-baby pair follow-up, safe disclosure, and returning with the baby for EID.

Take-away: Test, treat the same day with TLD, give co-trimoxazole instead of SP, track the VL every 3 months, and let the mother's VL and time on ART decide the baby's prophylaxis.

Case 2: Non-suppressed viral load in the third trimester

Grace, 31, P2+0 at 32 weeks, on TLD for 5 years. Her VL at booking (16 weeks) was 12,000 copies/mL; after 3 IAC sessions it is now 8,500 copies/mL. She attends Mengo ANC with a cough of 3 weeks.

1. What are the likely reasons for her non-suppression?

> In order: **poor adherence** (stigma, non-disclosure, alcohol, depression, pill fatigue, hyperemesis), **drug interactions** (rifampicin, iron/antacids taken with DTG, anticonvulsants), **malabsorption/vomiting**, and **drug resistance** (less common on DTG). Her cough raises **TB**, which also drives VL up and may involve rifampicin later.

2. What do you do about the cough?

> TB symptom screen is positive: **sputum GeneXpert**, TB LAM if CD4 below 200 or seriously ill, CXR with abdominal shielding. Check SpO_2 (PCP), temperature, weight trend. If TB confirmed, start RHZE by weight; add **DTG 50 mg** so DTG is **twice daily** throughout rifampicin and 2 weeks after. Screen household contacts.

3. How do you address the VL?

> Detailed adherence assessment (pill count, pharmacy refill records), home visit if needed, treatment supporter, **directly observed therapy**, mental health screen. Because she remains non-suppressed after IAC, present her to the **ART switch committee** for a regimen decision (resistance testing where available). Continue TLD meanwhile; never stop ART.

4. How does this affect delivery planning and the baby?

> Her infant will be **high-risk**. Plan facility delivery at a hospital; continue ART in labour; **avoid ARM, episiotomy, scalp electrodes, prolonged ROM**; CS only for obstetric indications (discuss with consultant if VL very high at term). Baby: **AZT + NVP for 6 weeks, then NVP for 6 weeks**, consider birth PCR if available, then routine EID. CTX from 6 weeks.

5. At 12 weeks postpartum her VL is 1,900 copies/mL. What now?

> She is still non-suppressed and breastfeeding, so **continue the infant's NVP** until she is suppressed or 4 weeks after breastfeeding stops. Continue IAC and act on the switch-committee decision. Repeat infant PCR if any symptoms. Keep the **mother-baby pair** closely linked.

Take-away: A non-suppressed VL is an adherence problem until proven otherwise; fix the barriers, look for TB and interactions, escalate to the switch committee, and treat the baby as high-risk.

Case 3: Unbooked mother, positive test in labour

Night shift at a HC IV in Mukono: Sarah, 19, G1, unbooked, arrives in active labour at 6 cm with intact membranes. Opt-out HIV testing in labour is positive (confirmed). She has never tested before.

1. What are your immediate actions for the mother?

> Brief, sensitive post-test counselling (full counselling can follow postpartum). **Start TLD now**. Continue labour care with the partograph. **Avoid ARM, routine episiotomy, scalp electrodes and unnecessary VEs**; avoid prolonged labour; active management of the third stage. Test for syphilis and HBsAg if not done.

2. Is caesarean section indicated?

> No. CS is for obstetric indications only. At 6 cm with normal progress, vaginal delivery with maternal ART and enhanced infant prophylaxis is appropriate. Unnecessary CS adds risks of sepsis and haemorrhage.

3. The baby is born weighing 3.2 kg. What prophylaxis, and when?

> **High-risk** (diagnosed in labour). **AZT 15 mg (1.5 mL) twice daily + NVP 15 mg (1.5 mL) once daily for 6 weeks**, then **NVP 20 mg once daily for 6 more weeks**. **Start within 6 hours**, do not wait for the morning. Birth PCR if available. Breastfeed within 1 hour, exclusively.

4. What follow-up do mother and baby need?

> Mother: baseline CD4, CrAg if CD4 200 or below, TB screen, CTX, **VL 3 months after ART start**, then every 3 months while breastfeeding, disclosure and partner testing. Baby: CTX from 6 weeks; **PCR at 6 weeks, 9 months, 6 weeks after weaning; antibody at 18 months**; extend NVP if the mother is still non-suppressed at 12 weeks postpartum.

> **Take-away:** Diagnosis in labour is an emergency for the baby: start maternal TLD immediately, protect the birth process, and give AZT + NVP within 6 hours.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.